বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা

BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Department of Computer Science & Engineering
Level- II Term-II
Midterm Examination, SPRING 2026
Course Code: CSE 223, Credit Hour-03
Course Title: Digital Logic Design
Marks: 90, Time: 1 hour 30 minutes

Notes:

a. Each question carries 30 marks.
b. Figure on the right of each question indicate marks for respective question.
c. **Scientific Calculator is not allowed.**
d. Required Postulates and Theorems of Boolean algebra are given in Table 1.

Course Learning Outcomes (CLO):

[CLO-1]: Explain the characteristics of Number system, Boolean algebra and different types of combinational and sequential circuits.
[CLO-2]: Apply Boolean, De Morgan's, and other related theorem to solve digital logic circuits.
[CLO-3]: Design various combinational and sequential circuits.

1. a. Explain the working principle of even and odd parity schemes, and describe how a parity bit is used to detect single-bit errors in data transmission. [CLO-1] — 10

b. Why complements are used in digital computers? Subtract $(1000100)_2 - (1010100)_2$ by using 1's complements. [CLO-1] — 10

OR

Convert the following decimal numbers into: [CLO-1]

- BCD
- Excess-3 code
- Gray code (for integer part only)

$$(126)_{10}$$

c. Simplify the following Boolean expression using Boolean algebra and verify using a truth table. [CLO-1] — 10

$$F = xy + xz' + yz$$

2. a. Why are NAND and NOR gates called universal gates? By using the truth table, Proof the De Morgan's theorem $(x + y)' = x'y'$. [CLO-2] 10

b. Find the sum of minterms and the product of maxterm for the following function, [CLO-2] 10

$$F(x, y, z) = 1$$

OR

Write short notes on the following terms: 5x2=10

i) Integrated circuit.

ii) Venn diagram.

c. Explain the concept of a register and discuss how registers are used for binary information processing. [CLO-2] 10

OR

What is the difference between canonical and standard form? Solve the following function using minimum no of literals. [CLO-2]

$$xy + x'z + yz$$

3. a. Discuss the role of don't-care conditions in Karnaugh Map simplification. Using K-map, obtain the simplified expressions for the following Boolean function: [CLO-1] 15

$$F = x'z' + y'z' + yz' + xyz$$

b. Using K-map, simplify the following truth table in i) sum of products and ii) product of sums. [CLO-1] 15

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1

0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Table 1:

1	Postulate 2	(a) $X+0 = X$	(b) $X.1 = X$
2	Postulate 5	(a) $X+X' = 1$	(b) $X. X' = 0$
3	Theorem 1	(a) $X+X = X$	(b) $X.X = X$
4	Theorem 2	(a) $X+1 = 1$	(b) $X.0 = 0$
5	Postulate 3, Commutative	(a) $X+Y = Y+X$	(b) $XY = YX$
6	Theorem 4, Associative	(a) $X+(Y+Z) = (X+Y)+Z$	(b) $X(YZ) = (XY)Z$
7	Postulate 4, Distributive	(a) $X(Y+Z) = XY+XZ$	(b) $X+YZ = (X+Y)(X+Z)$
8	Theorem 5, De-Morgan	(a) $(X + Y)' = X' Y'$	(b) $(XY)' = X'+Y'$
9	Theorem 6, Absorption	(a) $X+XY = X$	(b) $X(X+Y) = X$

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Department of Computer Science & Engineering
Level- II Term-II
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1. a. Compare BCD and Excess-3 code representations with suitable examples [CLO-1] — 10

   b. Perform the following conversions: — 2x5=10

      i. $753_8 \rightarrow (?)_2$
      ii. $A7F_{16} \rightarrow (?)_{10}$

      OR

      Perform binary subtraction using: — 2x5=10

      i. 1's Complement
      ii. 2's Complement

      $101101_2 - 011011_2$

      [CLO-1]

   c. Simplify the following Boolean function to a minimum number of literals. [CLO-1] — 10

      $F = A'B + A'BC + AB$

2. a. Why is NAND gate called a universal gate? Construct NOT, AND, and OR gates using only NAND gates. [CLO-2] 10

b. Verify De Morgan's theorem using a truth table: 10

$$(XY)'=X'+Y'$$

OR

Explain Half Adder and Full Adder with truth tables and logic diagrams. 10

c. Write short notes on: 10

i) Gray Code

ii) Error Detection using Parity Bit

[CLO-2]

OR

What is a register? Explain with a suitable figure how register transfer and binary information processing? 10

3. a. Explain the role of don't-care terms. Using K-map simplify: 15

$$F=x'z'+xy'+yz+x'yz$$

[CLO-1]

b. Using K-map simplify in SOP and POS: 15

$$F(A,B,C,D) = \sum_m(1,3,4,5,7,11,12,13)$$

[CLO-1]

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ii. 2's Complement

$$101101_2 - 011011_2$$

[CLO-1]

c. Simplify the following Boolean function to a minimum number of literals. [CLO-1] 10

$$F = A'B + A'BC + AB$$

2. a. Why is NAND gate called a universal gate? Construct NOT, AND, and OR gates using only NAND gates. [CLO-2] 10

b. Verify De Morgan's theorem using a truth table: 10

$$(XY)'=X'+Y'$$

OR

Explain Half Adder and Full Adder with truth tables and logic diagrams. 10

c. Write short notes on: 10

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$$F(A,B,C,D) = \sum_m(1,3,4,5,7,11,12,13)$$

[CLO-1]

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Spring 2026
Department of Computer Science and Engineering
Level-2 Term-II
Course Code: CSE 225
Course Title: Numerical Methods
Credit Hour: 3.0
Exam Duration: 1 hour 30 Minutes
Full Marks: 90

Notes:

a. Figure on the right of each question indicates the marks for the respective question.
b. Course Learning Outcomes are-

CO1: Recall the basics of accuracy and finite precision, round-off errors, truncation errors, and stability of numerical algorithms.

CO2: Apply appropriate numerical techniques and calculate the computational cost in the fields of science and engineering.

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1. **a.** A computer system records the value 0.00897654 with a four-digit mantissa. Calculate the round-off error introduced during storage. **[CO1→C3]** **[10]**

   b. Explain the concepts of overflow and underflow in floating-point operations with suitable examples. Demonstrate how these conditions occur in numerical methods. **[CO1→C3]** **[10]**

   c. Analyze the binary conversion of 6.56 and 1.75 using 8-bit representation. Perform binary addition and examine whether any precision loss occurs during reconversion to decimal. **[CO1→C4]** **[10]**

   OR

   An engineering experiment produces the following data:
   0.0045600 N, 345.600 J, 9.8700 kg, 0.75000 L, 1500.0 rpm, 6.78901 V

   i) Count the number of significant digits (SD) in each measurement.
   ii) Rank the measurements according to their precision. **[CO1→C4]**

2. **a.** Apply the False Position (Regula-Falsi) method to determine a root of the equation $2e^{x}sinx = 3$ within the interval [0,1], and compute the root correct to three decimal places. **[CO2→C3]** **[15]**

   b. Analyze the convergence of the Newton–Raphson method for the equation $x^3 - x - 1 = 0$ starting with the initial approximation $x_0 = 1$ and determine the root correct to three decimal places. **[CO2→C4]** **[15]**

   OR

   Analyze the application of the Newton–Raphson method to determine a real root of the equation $-4x + cosx + 2 = 0$ using the initial approximation $x_0 = 0.5$, and obtain the root correct to four decimal places. **[CO2→C4]**

3. **a.** Apply the Bisection method to find the real root of $x^3 - 6x + 4 = 0$ in the interval [0,1]. Compute the root accurate to three decimal places. **[CO2→C3]** **[15]**

b. Derive the Regular Falsi method formula using the equation of a straight line passing through two points on the curve. **[CO2→C3]** **[15]**

OR

Compare the convergence behavior of the Bisection method and False Position method for solving nonlinear equations. **[CO2→C4]**

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Spring-2026
Department of Computer Science and Engineering
Level-2, Term-2
Course Code: GED 221
Course Title: Engineering Economics
Credit Hour: 03
Exam Duration: 01 hour 30 Minutes
Full Marks: 90

Notes:

a. Answer all of the three questions.

b. The figure on the right of each question indicates marks for the respective question.

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1. a. Define Microeconomics. Explain the relationship between scarcity and choice. **[CO1]** **[15]**
   b. "Efficient allocation of resources is the key to solving the economic problem." Discuss. **[15]**

2. a. Define demand function. Using a suitable demand function, construct a demand schedule and illustrate the corresponding demand curve. **[15]**
   b. Define market equilibrium. Describe the process through which equilibrium price and quantity are determined in a competitive market. **[CO1]** **[15]**

OR

   a. Define supply. Compare individual supply from market supply. **[15]**
   b. Explain the law of supply with the help of a numerical example and figure. **[CO1]** **[15]**

3. a. Explain the injections and leakages in the circular flow of income model and discuss their main components. **[CO2]** **[15]**
   b. Describe the three-sector circular flow of income model with the help of a diagram and relevant equations. **[15]**

EXAMINATION CONFIDENTIAL

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Spring 2026

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE 221

Course Title: Computer Algorithms and Complexity

Credit Hour: 3.0

Exam Duration: 1 hour 30 Minutes

Full Marks: 90

Notes:

a. Figure on the right of each question indicates marks for respective question.

b. Course Learning outcomes are:

CO1: Analyze the asymptotic performance and write rigorous correctness proofs for algorithms.

CO2: Estimate the performance of various algorithms and data structures.

CO3: Apply important algorithmic design paradigms and methods analysis for problem solving.

CO4: Device efficient algorithms in common engineering design situations.

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1. a. Write down the pseudocode of magic square for odd order logic and generate a N x N magic square where N = 3. [CO1 → C3] **[10]**

b. Apply push, pop and front operation in a queue for the given array A = {10,3,4,5,6,7,8,3}. [CO1 → C3] **[10]**

c. Define the tree terminologies of a full binary tree and explain. [CO1 → C1] **[10]**

OR,

a. What is an algorithm? Briefly explain the essential characteristics of algorithm. [CO1 → C2] **[10]**

b. Construct a Step Table for the following algorithm [CO1 → C3] **[10]**

```
Algorithm Add(a, b,c, m, n)
{
   for i:=1 to m do
      for j:=1 to n do
         c[i, j] := a[i,j] + b[i,j];
}
```

c. Solve the following recurrence relations of the form [CO1 → C3] **[10]**

$$T(n) = \begin{cases} T(1), & n = 1 \\ aT\left(\frac{n}{b}\right) + f(n), & n > 1 \end{cases}$$

Consider the case in which a = 2 and b = 2, T (1) = 2 and f(n) = n.

2. a. Write a pseudo code for recursive search of a Binary Search Tree (BST). [CO2 → C3] **[10]**

b. Describe how a Min Heap is constructed from the following array [40, 30, 50, 22, 60, 55, 77, 35]. Show the steps. [CO2 → C4] **[10]**

c. Write a pseudocode for inserting a value into a Binary Search Tree (BST). Ensure that the pseudocode maintains the BST property: the left subtree contains values smaller than the node, and the right subtree contains values greater than the node. [CO2 → C3] **[10]**

3. a. Describe the step-by-step process of the Insertion Sort algorithm for sorting an array [8, 4, 6, 9, 2, 3, 1] in ascending order. [CO3 → C4] **[10]**

b. Write the pseudocode of the Merge Sort algorithm and explain how the Divide and Conquer approach is applied in this algorithm. [CO3 → C3] **[10]**

c. Given the array arr = {8, 3, 1, 7, 0, 10, 2}, demonstrate step-by-step how the Quick Sort algorithm sorts this array using the first element as the pivot. [CO3 → C4] **[10]**

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Spring 2026

07.06.26 (G.B)

Department of Computer Science and Engineering
Level- 2, Term-II
Course Code: MATH 221
Course Title: MATH-IV
Credit Hour: 03

Notes:

Time : 1 hr 30 mins
Full Marks: 90

a. Figure on the right of each question indicate marks for respective question.
b. Answer any **three (03) questions** out of the following **four (04)** questions including **Q1**.

1. a. The following frequency table gives the weekly number of hours worked including overtime by 70 workers of a factory. **Compute** arithmetic mean by: **[CO1]** **15**
i. Direct method
ii. Short-cut method

Number of hours worked	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Number of workers	5	12	15	25	8	3	2

b. In the winter season, the temperature (in Celsius) of the month of January in Cumilla is placed below. **Construct** the frequency distribution table of the temperature: 12°,14°,13°,13°,14°,13°,10°,10°,11°,12°,11°,10°,9°,8°,9°,11°,10°,10,8°,9°,7°,6°,6°,6°, 6°,7°,8°,9°,9°,7°,6°. **[CO1]** **07**

c. **Approximate** the arithmetic mean and geometric mean from the following set of observations: 2, 4, 8, 12, 16, 24. **[CO1]** **08**

2. a. A bag contains 4 Red marbles and 6 Blue marbles. A marble is drawn at random, its color is recorded, and it is put back into the bag. Then, a second marble is drawn. Draw the tree diagram for these two independent events. **Find** the probability that both marbles drawn are the same color? **[CO1]** **08**

b. A fire brigade has two fire engines operating independently. The probability that a specific fire engine is available when needed is 0.99. **Estimate** **08**
i) the probability that an engine is available when needed.
ii) the probability that neither is available when needed. **[CO1]**

c. Two ordinary six-sided dice were tossed. Set up a sample space for this experiment and hence **determine** the probability that: **14**
a) Sum of the numbers on the two dice is 7.

b) Numbers on die 1 are greater than the numbers on die 2.
c) First die shows an even number.
d) Numbers on both the dice are the same. **[CO1]**

3. a. In a community there are males and females are distributed as 2:3 of 100%. Suppose 7% of the males and 12% of the females are disabled. A person is chosen at random. If this person is female, then **compute** the probability that she is not disabled? **[CO2]** **07**

b. In a factory, three machines, **J** and **K** are used to make biscuits. Machine **J** makes 55% of the biscuits and the rest of the biscuits are made by machine **K**. It is known that 7% of the biscuits made by machine **J** are broken and 6% of the biscuits made by machine **K** are broken. Draw a tree diagram and **find** the probability that the biscuit is made by machine **K** and is not broken. **[CO2]** **09**

c. In the submission of an income tax return, a tax-payer has applied for the allowance of certain expenses, for which the legitimacy is subject to verification. A three-member review board operating on a rotation schedule receives applications for consideration. The first board, consisting mainly of members have vast experience, approves only 25% of such claims without verification. The second board, consisting mainly of midterm individuals, approves 50% of such claims. The third board with relatively newer members dubious about refuting claims approves 70% of such cases. There is a 20% chance that the application will be considered by the first board, a 50% chance that it will be considered by the second board, and a 30% chance that it will be considered by the third board. Now **Calculate** **14**
a) the probability that the application will be approved.
b) If the application is approved, what is the probability that it was considered by (i) the first board (ii) the second board (iii) the third board? **[CO2]**

4. a. An unbiased coin is tossed three times. Let **X** be the number of runs obtained as a result of the outcomes of this experiment. **Estimate** the probability distribution of **X**. **[CO2]** **08**

b. In a class of 180 students, all of whom took English and History, 15 failed History, 10 failed English and 5 failed both. **Find** the conditional probability that a student, who passed English also passed History. **[CO2]** **08**

c. A continuous random variable X has the following functional form: **14**

$$f(x) = \frac{2}{27}(1+x), \qquad 2 < x < 5$$
$$= 0, \qquad \textit{elsewhere}$$

(i) **Determine** it satisfies the condition $\int_{-\infty}^{\infty} f(x)\,dx = 1$

(ii) **Find** $P(3 < X < 4)$ and $P(X < 4)$. **[CO2]**

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Mid Term Examination, Spring 2026

Department of Computer Science and Engineering
Level- 2, Term-II
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Course Title: MATH-IV
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d) Numbers on both the dice are the same. **[CO1]**

3. a. In a community there are males and females are distributed as 2:3 of 100%. Suppose 7% of the males and 12% of the females are disabled. A person is chosen at random. If this person is female, then **compute** the probability that she is not disabled? **[CO2]** **07**

b. In a factory, three machines, **J** and **K** are used to make biscuits. Machine **J** makes 55% of the biscuits and the rest of the biscuits are made by machine **K**. It is known that 7% of the biscuits made by machine **J** are broken and 6% of the biscuits made by machine **K** are broken. Draw a tree diagram and **find** the probability that the biscuit is made by machine **K** and is not broken. **[CO2]** **09**

c. In the submission of an income tax return, a tax-payer has applied for the allowance of certain expenses, for which the legitimacy is subject to verification. A three-member review board operating on a rotation schedule receives applications for consideration. The first board, consisting mainly of members have vast experience, approves only 25% of such claims without verification. The second board, consisting mainly of midterm individuals, approves 50% of such claims. The third board with relatively newer members dubious about refuting claims approves 70% of such cases. There is a 20% chance that the application will be considered by the first board, a 50% chance that it will be considered by the second board, and a 30% chance that it will be considered by the third board. Now **Calculate** **14**
a) the probability that the application will be approved.
b) If the application is approved, what is the probability that it was considered by (i) the first board (ii) the second board (iii) the third board? **[CO2]**

4. a. An unbiased coin is tossed three times. Let **X** be the number of runs obtained as a result of the outcomes of this experiment. **Estimate** the probability distribution of **X**. **[CO2]** **08**

b. In a class of 180 students, all of whom took English and History, 15 failed History, 10 failed English and 5 failed both. **Find** the conditional probability that a student, who passed English also passed History. **[CO2]** **08**

c. A continuous random variable X has the following functional form: **14**

$$f(x) = \frac{2}{27}(1+x), \qquad 2 < x < 5$$
$$= 0, \qquad \textit{elsewhere}$$

(i) **Determine** it satisfies the condition $\int_{-\infty}^{\infty} f(x)\,dx = 1$

(ii) **Find** $P(3 < X < 4)$ and $P(X < 4)$. **[CO2]**

Examination Confidential

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি (বাইউস্ট), কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Spring 2026

Department of Computer Science and Engineering
Level- 2, Term-II
Course Code: MATH 221
Course Title: MATH-IV
Credit Hour: 03

Notes:

Time : 1 hr 30 mins

Full Marks: 90

a. Figure on the right of each question indicate marks for respective question.
b. Answer any **three (03) questions** out of the following **four (04)** questions including **Q1**.

1. a. The following frequency table gives the weekly number of hours worked including overtime by 70 workers of a factory. **Compute** arithmetic mean by: **[CO1]** — **15**
i. Direct method
ii. Short-cut method

Number of hours worked	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Number of workers	5	12	15	25	8	3	2

b. In the winter season, the temperature (in Celsius) of the month of January in Cumilla is placed below. **Construct** the frequency distribution table of the temperature: 12°,14°,13°,13°,14°,13°,10°,10°,11°,12°,11°,10°,9°,8°,9°,11°,10°,10,8°,9°,7°,6°,6°,6°, 6°,7°,8°,9°,9°,7°,6°. **[CO1]** — **07**

c. **Approximate** the arithmetic mean and geometric mean from the following set of observations: 2, 4, 8, 12, 16, 24. **[CO1]** — **08**

2. a. A bag contains 4 Red marbles and 6 Blue marbles. A marble is drawn at random, its color is recorded, and it is put back into the bag. Then, a second marble is drawn. Draw the tree diagram for these two independent events. **Find** the probability that both marbles drawn are the same color? **[CO1]** — **08**

b. A fire brigade has two fire engines operating independently. The probability that a specific fire engine is available when needed is 0.99. **Estimate** — **08**
i) the probability that an engine is available when needed.
ii) the probability that neither is available when needed. **[CO1]**

c. Two ordinary six-sided dice were tossed. Set up a sample space for this experiment and hence **determine** the probability that: — **14**
a) Sum of the numbers on the two dice is 7.

b) Numbers on die 1 are greater than the numbers on die 2.
c) First die shows an even number.
d) Numbers on both the dice are the same. **[CO1]**

3. a. In a community there are males and females are distributed as 2:3 of 100%. Suppose 7% of the males and 12% of the females are disabled. A person is chosen at random. If this person is female, then **compute** the probability that she is not disabled? **[CO2]** **07**

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